When Does Recurrent Memory Help? Six Failure Modes and a Methodology for Honest Evaluation

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Abstract

We began with a simple question: why did a recurrent memory module sometimes look useful, sometimes invisible, and sometimes actively harmful on the same synthetic callback task? In a 16-version campaign on Memory Residuals, a slot-attention memory module attached to Qwen3-0.6B and Qwen3-1.7B, the answer was not a single bug. It was an evaluation problem. Token-averaged chain evaluation diluted a localized callback gain by roughly two orders of magnitude; frozen and joint training measured different causal pathways; the D4v2 callback template leaked a 32-way category prior; more than half of the D4v2 callback windows exposed evidence directly to the LM path; evidence redaction had to be audited before a zero lift could be trusted; and a random-ID corpus removed the prior but exposed a writer objective that still did not force chain-specific bindings into M_{c}. We turn these failures into a methodology: callback-aware evaluation, evidence- redacted memory floors, empirical and base-model prior audits, window-leakage counts, and a test-time readout audit that asks whether the writer encoded the answer at all. The result is deliberately negative but actionable: most apparent memory gains on D4v2 are shortcuts, while D5 shows that removing shortcuts is not sufficient without writer-side extractive supervision. We close with an F2 prescription: NIAH-style writer warmup that gives M_{c} a direct chain-specific training signal before the LM head can learn a shortcut.

1 Introduction

Recurrent memory is attractive because it promises a bounded-cost state that survives across sessions. In a conversational assistant, that state should remember a user's code, preference, or updated fact even when the evidence is no longer in the language model's local attention window. The hard question is not whether an augmented model can reduce loss somewhere. It is whether the recurrent state is the causal path that carries the chain-specific binding.

This paper is an empirical post-mortem of that question. We study Memory Residuals: a fixed-size memory state M_{c} written once per session and read back into a Qwen3 backbone through a depth-routed attention residual. Across versions v11–v16, the system produced encouraging numbers, contradictory numbers, and finally clean negative numbers. The campaign changed our interpretation of the result. The main contribution is not a state-of-the-art memory model; it is a set of failure modes and diagnostics that determine when a memory result is honest.

The Running Task. Most of the campaign uses synthetic multi-session chains. An evidence session states a binding; later, a callback session asks for it. The model is scored on the answer tokens. In D4v2 the binding is a closed-set persona fact such as "favorite color." In D5 the answer is a per-chain random alphanumeric code and the evidence-session LM loss is masked so the callback span is the

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only supervised answer location. The intended invariant is simple: if the evidence is outside the LM window and the callback does not reveal the answer, only M_c should carry the value.

What Went Wrong. Five concrete findings forced a reframe. First, the standard `eval_chain.py` metric averages over full score windows. On v14k, callback-aware evaluation reports $\Delta^{\text{cb}}_{\text{nm}-\text{m}} = +1.44$ nats, but standard full-window evaluation reports $\Delta_{\text{nm}-\text{m}} \approx 0$ because the callback answer tokens are only about 1.4% of the scored window. Second, the frozen-backbone and joint-training regimes measure different causal systems: the frozen 0.6B path can show apparent memory help, while the joint 0.6B path learns the D4v2 prior directly and collapses Δ_{evidence} to zero. Third, D4v2 leaks a category prior: the callback "Your favorite COLOR is __" reduces the answer to a 32-item closed set, and the empirical prior has CE 2.91 nats per callback-mask token. v15b reaches 2.97 nats, within 0.06 nats of that prior. Fourth, D5 removes that closed-set cue but does not cure the writer objective: v16a still has near-zero Δ_{evidence} because the writer is not forced to encode the chain-specific code. Fifth, the evidence-redacted floor is clean: Audit A2 shows default, cross-chain, skip, and zero redactions agree within 0.05 nats, so a zero lift is not an artifact of leaky redaction.

Contributions. We contribute four evaluation tools and one prescription.

1. **Callback-aware standalone evaluation.** `tools/eval_callback.py` scores only annotated callback answer tokens and reports an evidence- redacted M_c floor.
2. **Multi-axis leakage audits.** We measure empirical $P(\text{item} \mid \text{category})$ CE, frozen-base callback CE, window-leakage rates, and redaction validity.
3. **D5 random-ID stress test.** The D5 corpus removes the D4v2 category prior and masks evidence-session LM loss, exposing whether the writer objective alone stores the binding.
4. **TTT-readout audit.** A planned diagnostic freezes the model and optimizes only readout-side parameters on callback tokens to test whether the writer encoded the answer in M_c.
5. **F2 prescription.** We propose NIAH-style writer warmup: a probe head gives direct $M_c \rightarrow$ answer supervision before the LM-head path is allowed to dominate.

2 Memory Residuals Architecture

Memory Residuals augments a pretrained causal LM with a recurrent state $M_c \in \mathbb{R}^{K \times d}$, where $K = 128$ slots and d is the backbone hidden size. The state is updated at session boundaries, not per token. The architecture has three stages: extraction from the completed session, a judge/write step that updates M_c, and a readout that injects m_t into the next session's residual stream.

Extract. Given a session representation $C_t \in \mathbb{R}^{N_t \times d}$, a learned query bank M_{in} cross-attends to produce a candidate memory:

$$M_{\text{new}} = \text{softmax}\left(\frac{M_{\text{in}} W_Q^{\text{ext}} (C_t W_K^{\text{ext}})^\top}{\sqrt{d}}\right) C_t W_V^{\text{ext}}. \tag{1}$$

The v15 recipe uses a contextual extraction source (`hidden_14`) and an input RMSNorm before extraction. That norm was introduced after gradient audits showed unnormalized hidden states could drive extract-key gradients roughly $50\times$ too large.

Judge And Write. The candidate M_{new} competes with the previous state through a slot-attention writer:

$$P = [M_c^{t-1} \parallel M_{\text{new}}], \qquad M_c^t = \text{Judge}_\theta(P). \tag{2}$$

Earlier versions used a cross-attention judge and later versions use a slot-attention writer with orthogonal query initialization and slot position embeddings. The important empirical point is that the writer can settle into a content-blind fixed point: M_c becomes useful as a generic "memory is on" prior while carrying little chain-specific content.

Table 1: Failure modes and diagnostics. "Fix" means the methodology we adopt in this paper, not a claim that the architecture is solved.

Name	Symptom	Diagnostic	Fix
Token-averaged dilution	Standard eval reports $\Delta_{\mathrm{nm-m}} \approx 0$ although callback tokens improve.	Callback-token-only CE; compare score-token counts.	Use `eval_callback.py`.
Joint-training invisibility	Unfrozen backbone collapses $\Delta_{\mathrm{evidence}}$ to ≈ 0.	Freeze-vs-joint ablation; compare to empirical prior.	Report memory-disabled and prior baselines.
Category-cue leak	"favorite COLOR" reduces answer to a 32-way prior.	$P(\mathrm{item} \mid \mathrm{category})$ CE = 2.91 nats; v15b CE = 2.97.	Random IDs; remove category cue.
Window leakage	Evidence directly enters the LM attention window.	Count evidence positions in callback windows.	Use smaller callback window or forced evidence gap.
Redaction uncertainty	Zero lift could be dismissed as a leaky floor.	Default/cross/skip/zero redaction agreement within 0.05 nats.	Keep evidence-redacted floors and audit them.
Writer-objective failure	D5 removes priors but $\Delta_{\mathrm{evidence}}$ remains near zero.	D5 run plus TTT-readout/probe audits.	F2 NIAH writer warmup.

Five pathways from M_{c} or context to the LM head.

1. Intended path: evidence session $\rightarrow C_t \rightarrow$ writer $\rightarrow M_{\mathrm{c}} \rightarrow m_t \rightarrow$ LM head.
2. Content-blind memory prior: writer emits a similar M_{c} on every chain; readout shifts the LM toward the callback marginal.
3. Backbone template prior: unfrozen LM learns $P(\mathrm{item} \mid \mathrm{category})$ from callback tokens.
4. Window path: evidence remains in the local attention window, bypassing M_{c}.
5. Measurement path: token-averaged eval or unaudited redaction changes the apparent contribution of memory.

Figure 1: Causal pathways that must be separated before a memory result can be interpreted. Only the first path is chain-specific recurrent memory.

Read. When scoring the next session, token representations X attend to M_{c}:

$$m_t = \mathrm{softmax}\left(\frac{XW_Q^{\mathrm{r}}(M_{\mathrm{c}}W_K^{\mathrm{r}})^\top}{\sqrt{d}}\right) M_{\mathrm{c}}W_V^{\mathrm{r}}. \tag{3}$$

The readout m_t has the same shape as a residual-stream tensor and is registered as an extra source in a Block-Attention Residual router. A learned routing mass α_{mem} determines how much the LM computation uses the memory source at each routed sublayer. This design is intended to be additive on top of the backbone, but the experiments below show that "additive" is not enough: every non-memory route to the callback value must be audited.

3 Failure Modes At A Glance

Table 1 summarizes the six failure modes surfaced by the campaign. The first four can make memory look effective when it is not, or hide a localized effect when it is real. The fifth prevents a false dismissal of the metric. The sixth is the remaining architecture/training problem after the shortcut paths are removed.

4 Failure Modes 1–2: Dilution And Joint-Training Invisibility

Callback Benefit Is Localized. The first apparent contradiction was v14k. The in-trainer and callback-aware metrics showed a large callback-token benefit, while standard chain evaluation made

Table 2: D4v2 ablation and audit results. CE/prior rows are from the A3 template-prior audit on 128 validation chains unless noted. Lift values mix in-trainer and post-hoc audits where indicated; missing trajectory details are left as TODOs rather than interpolated.

Run	Regime	$\mathrm{CE}_{\mathrm{mem}}$	Gap to 2.91	$\Delta^{\mathrm{cb}}_{\mathrm{nm-m}}$	$\Delta_{\mathrm{evidence}}$	Readin
v14k	0.6B frozen	4.98	+2.07	+1.44	+0.071	Large gain; 9 tion.
v15a	0.6B frozen	5.15	+2.24	+1.33	-0.043	Reproc hoc flo idence.
v15b	0.6B joint	2.97	+0.06	+0.007	-0.002	Matche memor ishes.
v15c	0.6B frozen, embed extract	**[TODO: NUMBER]**	**[TODO: NUMBER]**	**[TODO: NUMBER]**	+0.005–+0.020	Hidder better.
v15e	1.7B frozen	4.76	+1.85	**[TODO: NUMBER]**	+0.066 / -0.09	Strong unstab lift.
v15f	1.7B joint	3.21	+0.30	**[TODO: NUMBER]**	≈+0.04	Conver same collaps

the same checkpoint look neutral or worse. The callback-aware post-hoc eval on 200 D4v2 validation chains reports

$$\mathrm{CE}_{\mathrm{mem}} = 4.98, \quad \mathrm{CE}_{\mathrm{nomem}} = 6.42, \quad \mathrm{CE}_{\mathrm{shuffle}} = 5.04, \quad \mathrm{CE}_{\mathrm{floor}} = 5.05.$$

Thus $\Delta^{\mathrm{cb}}_{\mathrm{nm-m}} = +1.44$, $\Delta^{\mathrm{cb}}_{\mathrm{sh-m}} = +0.054$, and $\Delta_{\mathrm{evidence}} = +0.071$. The full-window standard eval reports $\Delta_{\mathrm{nm-m}} = -0.085$ on the same checkpoint. These are not inconsistent: the callback mask is a tiny part of the score window, and memory can add small noise on non-callback tokens that dominates a full-window average.

Most Of The Apparent Benefit Survives Redaction. The same v14k number also reveals a second problem. The evidence- redacted floor still gives $\Delta_{\mathrm{nm-m,floor}} = +1.37$. Therefore $1.37/1.44 = 95.1\%$ of the apparent no-memory vs memory gain survives when M_{c} is built without the answer. The honest chain-specific quantity is not $\Delta_{\mathrm{nm-m}}$ but $\Delta_{\mathrm{evidence}} = \Delta_{\mathrm{nm-m}} - \Delta_{\mathrm{nm-m,floor}}$.

Joint Training Changes The Object Of Measurement. The v15 matrix makes the point sharper. With a frozen backbone, the memory path can change callback CE because the LM head cannot simply fit the D4v2 closed-set marginal. With an unfrozen backbone, the LM path can learn the callback distribution directly. v15b, the 0.6B joint-trained run, reaches $\mathrm{CE}_{\mathrm{mem}} \approx 2.97$ nats and $\Delta_{\mathrm{evidence}} \approx 0$. This is not evidence that memory is unnecessary in general; it is evidence that the benchmark allowed a non-memory path to solve the proxy objective.

5 Failure Mode 3: The Category-Cue Corpus Leak

D4v2 was meant to test whether M_{c} stores a persona binding. Its callback template inadvertently answers a different question: can the model learn a category-conditional closed-set prior? The callback repeats the category in the user question and immediately before the answer:

> User: Quick question, what was my favorite COLOR again?
> Assistant: Your favorite COLOR is RED.

Given the category, the answer space is roughly 32 items. Audit A3 builds a Laplace-smoothed empirical $P(\mathrm{item} \mid \mathrm{category})$ from the 5000-chain training set and scores 128 validation callbacks. The resulting prior CE is 2.9087 nats per callback-mask token. The joint 0.6B v15b checkpoint scores 2.9655 nats at best and 2.9394 nats at final, within 0.06 and 0.03 nats of the prior. That is the smoking gun: the unfrozen backbone has learned the template prior, not retrieval.

Table 3: D4v2 versus D5 corpus comparison. D5 removes the category cue and uses per-chain random alphanumeric codes; observed v16a results show that removing the prior exposes, but does not solve, the writer objective problem.

Corpus	Prior / floor CE	Base CE	Observe
D4v2 closed-set persona	2.91	7.69 / 8.76	v14k +0.07
D5 random codes	$\sim$10 nats per ID-character; first observed callback CE $\sim$7	**[TODO: NUMBER]**	v16a -0.002 at step 100;

Frozen-base controls rule out the opposite explanation that Qwen3 already knew this prior. Without memory or fine-tuning, callback-only CE is 7.69 nats for Qwen3-0.6B and 8.76 nats for Qwen3-1.7B on the same masked tokens. The prior is learned during the chain training regime.

6 Failure Mode 4: Random IDs Are Not A Cure

D5 implements the obvious fix. It replaces closed-set persona items with per-chain random alphanumeric codes and removes the category cue. The evidence session acknowledges the code without echoing it in the assistant response, and `-mask_evidence_session_loss` zeros LM loss over evidence sessions. In principle, the callback answer has no useful non-memory path: the code is random, the callback does not identify a category, and the LM head is not directly trained on the evidence answer span.

The first D5 result is therefore important precisely because it is negative. v16a, a frozen 0.6B D5 run, reports $\text{evidence_lift} = -0.0024$ at step 100; the v17 proposal packet summarizes the later state as $\text{evidence_lift} = +0.011$ with $\Delta^{\text{cb}}_{\text{nm}-\text{m}} = -0.076$, InfoNCE loss $1.375 \approx \log 4$, D3-MC pair/self 0.012, and $\alpha_{\text{mem}} \approx 0.01$. Removing the prior made the evaluation honest, but it did not create a training signal that forces the writer to bind the code into M_{c}.

This is a different failure from the D4v2 category leak. On D4v2 the LM path can win by learning a closed-set marginal. On D5 the marginal is unhelpful, but the writer still receives its gradient through a long, weak path: callback NLL $\rightarrow$ LM head $\rightarrow$ depth-router $\rightarrow$ readout $\rightarrow$ M_{c} $\rightarrow$ writer. Near random initialization, wrong codes and uninformative memories look similarly bad to the LM loss. The writer has no direct extractive objective saying, "the code in this evidence session must be present in M_{c}." That is the motivation for F2.

7 Failure Mode 5: Leak Audits As Methodology

An honest negative result requires proving that the diagnostic itself is not broken. Audit A2 traces both the in-trainer floor and standalone callback floor. The in-trainer floor builds M_{c} fresh from a single non-evidence session in the same chain; the standalone floor walks the prefix while replacing evidence sessions with distractors. Neither path carries a cached rolling state across evaluation passes.

The adversarial redaction test is stronger than the code trace. On v15a/best, the default redaction gives $\text{CE}_{\text{floor}} = 5.1235$; cross-chain redaction gives 5.0911; skip-evidence gives 5.0857; zeroed-input redaction gives 5.0896. The spread is 0.038 nats, inside the pre-declared 0.05-nat band, and the direction is opposite a leak: the default floor is slightly worse, not better, than cross/zero. On v15b/best, all floor variants lie within 0.008 nats. Thus $\Delta_{\text{evidence}} \approx 0$ is not a redaction artifact.

Audit A1 adds a separate lesson. At D4v2's v15 default `window_k=4`, 58.5% of train chains and 59.6% of validation chains have at least one evidence session inside the LM attention window at callback scoring time. At `window_k=3`, the leak is still 41.4% / 43.4%. Only `window_k=1` eliminates direct evidence visibility. This makes callback-window counts a required part of future corpus reports.

Run	Corpus	Scale/regime	$\Delta^{\mathrm{cb}}_{\mathrm{nm-m}}$	$\Delta_{\mathrm{evidence}}$
v14k	D4v2	0.6B frozen	+1.44	+0.071
v15a	D4v2	0.6B frozen	+1.33	-0.043 (post-hoc floor)
v15b	D4v2	0.6B joint	+0.007	-0.002
v15e	D4v2	1.7B frozen	+2.03 (trainer)	+0.066 / -0.09
v15f	D4v2	1.7B joint	**[TODO: NUMBER]**	≈+0.04
v16a	D5	0.6B frozen	-0.076	+0.011
v16b	D5	1.7B frozen	**[TODO: NUMBER]**	**[TODO: NUMBER]**

Figure 2: Trajectory summary for the main phase-aligned callback metric. The underlying logs are heterogeneous; this figure therefore shows audited checkpoints rather than an interpolated curve.

Table 4: D5 TTT-readout audit placeholder. The parent run will fill these values when `tools/d5_ttt_readout.py` completes. All unknown numbers are deliberately left as `\todo`.

Checkpoint	Probe params trained	Base CE	TTT CE	TTT $\Delta_{\mathrm{evidence}}$	Verdict
v15a/best	W_Q, W_K, W_V only	**[TODO: NUMBER]**	**[TODO: NUMBER]**	**[TODO: NUMBER]**	**[TODO: writer vs reado bottleneck]**
v15e/best	W_Q, W_K, W_V only	**[TODO: NUMBER]**	**[TODO: NUMBER]**	**[TODO: NUMBER]**	**[TODO: writer vs reado bottleneck]**

8 Prescription: F2 NIAH Writer Warmup

The F2 prescription is to add a writer-side extractive objective before the LM head is allowed to use memory. The proposed v17a/v17e warmup uses the D5 random-code corpus and a probe head that reads M_{c} and predicts the code. The probe is discarded at evaluation time; it is a training scaffold, not an inference path. If the final LM-head callback evaluation improves after the probe is removed, the result means the writer has learned to store content that the normal readout can use.

The warmup tests two hypotheses cleanly. If probe CE drops well below the D5 marginal but downstream $\Delta_{\mathrm{evidence}}$ remains near zero, then the writer can encode the answer and the bottleneck is readout/router use. If probe CE stays at the marginal, the writer cannot encode the binding even under direct supervision and the architecture itself needs a writer-side change. Either outcome is more informative than another LM-only training run.

9 Related Work

Our findings align with a broad lesson from long-context and memory evaluation: shortcuts are the default when the benchmark permits them. NoLiMa removes literal overlap between query and evidence and shows that ordinary long-context retrieval performance can be template-driven rather than robust memory use [1]. RULER argues that vanilla needle-in-a-haystack is saturated and adds UUIDs, multiple needles, variable tracking, aggregation, and hard distractors to expose weaker retrieval behavior [2]. Lost in the Middle shows position and recency/primacy effects even when context length is nominally large [3]. LongBench includes no-context checks to expose parametric priors [4]; our callback-only and empirical-prior audits are the memory analogue. LongMemEval separates indexing, retrieval, and reading, motivating our separation of write, readout, and decode diagnostics [6].

Memory architectures face the same credit-assignment issue. Recurrent Memory Transformer, Associative RMT, AutoCompressors, Memorizing Transformers, TRIME, Block-Recurrent Transformers, and classical differentiable memory systems all introduce pathways that can, in principle, carry long-range information [9; 10; 11; 8; 12; 13; 14]. Our results emphasize that adding a pathway is not the same as proving the model used it. The evaluation must include memory-off, memory-shuffled, evidence-redacted, prior-only, and window-leakage controls.

Run	Scale	Warmup/probe CE	$\Delta^{\mathrm{cb}}_{\mathrm{nm-m}}$	$\Delta_{\mathrm{evidence}}$
v17a	0.6B	**[TODO: NUMBER]**	**[TODO: NUMBER]**	**[TODO: NUMBER]**
v17e	1.7B	**[TODO: NUMBER]**	**[TODO: NUMBER]**	**[TODO: NUMBER]**

Figure 3: Placeholder for v17 F2 trajectories. Positive $\Delta_{\mathrm{evidence}}$ after probe removal would close the loop: D5 removes the shortcut and F2 supplies the missing writer gradient.

10 Conclusion And Open Questions

The campaign gives a conservative answer to "when does recurrent memory help?" It helps only when the task and evaluator make the non-memory paths uninformative and when the writer receives a gradient that can actually encode the binding. D4v2 failed the first condition: category cues, window leakage, and token-averaged eval made the memory result ambiguous. D5 satisfies the first condition better and exposes the second: LM-only callback loss is too weak to train the writer out of a content-blind fixed point.

The next decisive experiment is F2. A positive v17a/v17e result would show that direct writer supervision plus shortcut-free data yields chain-specific memory under the normal LM-head readout. A negative result would be equally useful: it would localize the failure to the readout/router or the memory representation itself. Either way, the methodological bar is now clear: report the shortcuts before reporting the memory gain.

Reproducibility Statement

All numbers in the main text are drawn from the run ledger and audits in `results/exp2_chain_recipe/` or from callback-aware JSON outputs in `results/eval_v14v15/`. Missing values are marked with **[TODO: NUMBER]**. The relevant tools are `tools/eval_callback.py`, `tools/audit_a1_window_leakage.py`, `tools/audit_a2_redaction.py`, `tools/audit_a3_template_prior.py`, `tools/build_synthetic_random_codes.py`, and the planned `tools/d5_ttt_readout.py`.

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A Audit Details

A.1 Window Leakage Counts

For D4v2, evidence positions are sampled from body positions $0, \ldots, 8$ and the callback is at position 9. Audit A1 counts whether an evidence session falls in the callback scoring window. At `window_k=4`, 2926/5000 train chains and 298/500 validation chains leak at least one evidence session. At `window_k=3`, the corresponding counts are 2068/5000 and 217/500. At `window_k=2`, they are 1088/5000 and 114/500. Only `window_k=1` eliminates this direct path.

A.2 Redaction Variants

Audit A2 evaluates five floor variants: original evidence (match), default distractor-pool redaction, cross-chain non-evidence redaction, skip-evidence, and zeroed-token evidence. On v15a/best, the redaction-style spread is 0.038 nats; on v15b/best it is 0.0066 nats. The default floor is slightly worse than cross/zero, so the default is not hiding evidence in the dangerous direction.

A.3 Template Prior Construction

Audit A3 parses $(\mathrm{category}, \mathrm{item})$ from D4v2 chain names and builds an add-half-smoothed empirical prior over each category. The train counts range from 593 to 652 chains per category. The validation prior CE over the first 128 chains is 2.9087 nats per callback-mask token. The per-item mean is essentially $\log 32$; the per-token mean is lower because 40/128 validation items are multi-token.

B Callback-Aware Metric Definition

Let S_c be the callback session of chain c and A_c its annotated answer-token mask. For a memory construction r, define

$$\mathrm{CE}_r^{\mathrm{cb}}(c) = \frac{1}{|A_c|} \sum_{i \in A_c} -\log p_\theta(x_i \mid x_{<i}, M_{\mathrm{c},r}(c)).$$

The reported callback deltas are

$$\Delta_{\mathrm{nm-m}}^{\mathrm{cb}} = \mathrm{CE}_{\mathrm{nomem}}^{\mathrm{cb}} - \mathrm{CE}_{\mathrm{mem}}^{\mathrm{cb}}, \qquad \Delta_{\mathrm{sh-m}}^{\mathrm{cb}} = \mathrm{CE}_{\mathrm{shuffle}}^{\mathrm{cb}} - \mathrm{CE}_{\mathrm{mem}}^{\mathrm{cb}},$$

and

$$\Delta_{\mathrm{evidence}} = \Delta_{\mathrm{nm-m}}^{\mathrm{cb}} - \left(\mathrm{CE}_{\mathrm{nomem,floor}}^{\mathrm{cb}} - \mathrm{CE}_{\mathrm{mem,floor}}^{\mathrm{cb}}\right).$$

Positive $\Delta_{\mathrm{evidence}}$ means evidence-bearing M_{c} helps more than evidence-redacted M_{c}.

C Additional Tables

D Limitations

The paper is a first draft under an active experiment campaign. Several v17 and TTT-readout numbers are intentionally left as placeholders. The D5 corpus removes the most obvious D4v2 prior but still

Table 5: Base callback CE without memory or fine-tuning on D4v2 callback-mask tokens.

Backbone	CE on callback tokens
Qwen3-0.6B base	7.686
Qwen3-1.7B base	8.764
Uniform over 256 closed-set items	5.545
Uniform within cued 32-item category	3.466
Empirical per-token template prior	2.909

Table 6: A2 redaction audit summary.

Checkpoint	Match $\Delta_{\mathrm{nm-m}}$	Default $\Delta_{\mathrm{evidence}}$	Cross $\Delta_{\mathrm{evidence}}$	Redaction spread
v15a/best	+1.3377	-0.0310	-0.0634	0.038
v15b/best	+0.0074	-0.0021	-0.0078	0.0066

uses synthetic templates; future natural-memory benchmarks such as LoCoMo and MSC should be reported with the same no-memory, prior, redaction, and shuffle controls. Finally, Memory Residuals itself is only one architecture. The central claim of this paper is methodological: without these controls, any recurrent-memory architecture can appear to work through hidden pathways.